Authors Response and corrections

Dear Editor,

Thank you for giving me opportunity to submit a revised draft of my manuscript titled **"Microwave Filter for SAIH antenna Sensor in Insulin Delivery System to Prevent Pancreas Thermogenesis Noise Signal Using TQWT Transform"** to **Biomedical Signal Processing and Control"**

Most of the comments and suggestions made by the reviewers have been incorporated into the final version of the work. In the version of the manuscript that we have had revised, the places where changes were made have been marked. The following is an in-depth response to each individual reviewer's questions and remarks, which can be seen below.

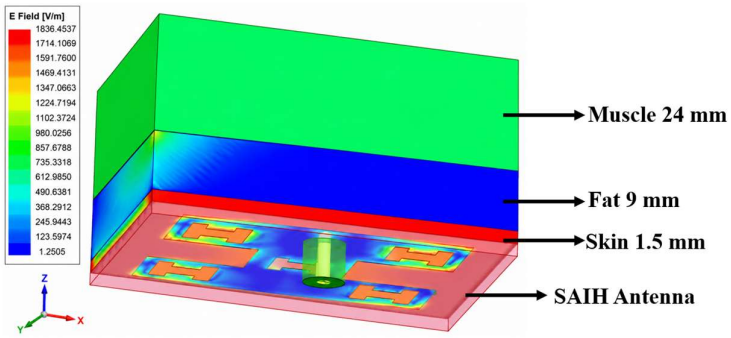
<table>
<tr><th colspan="2">Reviewer</th><th>Page No.</th></tr>
<tr><td>Comment.1</td><td>The claimed pancreas-targeted sensing capability at 4.2 GHz appears overly optimistic considering the severe attenuation and scattering in multilayer abdominal tissues. More realistic multilayer human-body modeling, SAR analysis, and penetration-depth validation should be included to support the feasibility of deep-tissue glucose monitoring.</td><td></td></tr>
<tr><td>Response.1</td><td>


Fig.1: E-field Distribution
Proposed antenna operates at 4.2 GHz, with skin, fat, and muscle represented as separate layers. Fig.1 (Supplementary file) shows the E-filed distribution of skin, fat, and muscle and table 21 shows the SAR values for a multilayer tissue phantom
Table 21: SAR values for a multilayer tissue phantom
<table>
<tr><th rowspan="2">Case</th><th>Skin (mm)</th><th>Fat (mm)</th><th>Muscle (mm)</th><th rowspan="2">Total thickness (mm)</th><th>Skin SAR (W/kg)</th><th>Fat SAR (W/kg)</th><th>Muscle SAR (W/kg)</th></tr>
<tr><th colspan="3">Ultrasound</th><th colspan="3">SAIH Antenna</th></tr>
<tr><td>1</td><td>1.0</td><td>5</td><td>15</td><td>21.0</td><td>0.80</td><td>0.25</td><td>0.12</td></tr>
<tr><td>2</td><td>1.0</td><td>9</td><td>24</td><td>34.0</td><td>0.72</td><td>0.21</td><td>0.09</td></tr>
<tr><td>3</td><td>1.5</td><td>5</td><td>20</td><td>26.5</td><td>0.95</td><td>0.28</td><td>0.13</td></tr>
<tr><td>4</td><td>1.5</td><td>9</td><td>24</td><td>34.5</td><td>0.86</td><td>0.23</td><td>0.10</td></tr>
<tr><td>5</td><td>1.5</td><td>12</td><td>24</td><td>37.5</td><td>0.82</td><td>0.19</td><td>0.08</td></tr>
<tr><td>6</td><td>2.0</td><td>5</td><td>24</td><td>31.0</td><td>1.05</td><td>0.30</td><td>0.14</td></tr>
<tr><td>7</td><td>2.0</td><td>9</td><td>20</td><td>31.0</td><td>0.98</td><td>0.25</td><td>0.12</td></tr>
<tr><td>8</td><td>2.0</td><td>9</td><td>30</td><td>41.0</td><td>0.91</td><td>0.22</td><td>0.08</td></tr>
<tr><td>9</td><td>2.5</td><td>12</td><td>30</td><td>44.5</td><td>1.08</td><td>0.20</td><td>0.07</td></tr>
<tr><td>10</td><td>3.0</td><td>15</td><td>30</td><td>48.0</td><td>1.20</td><td>0.18</td><td>0.06</td></tr>
</table>
</td><td>22</td></tr>
</table>

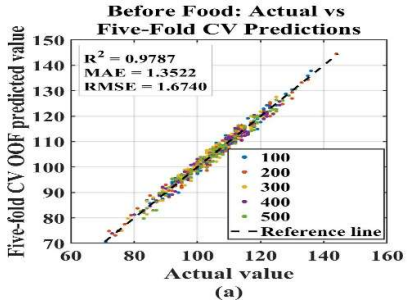
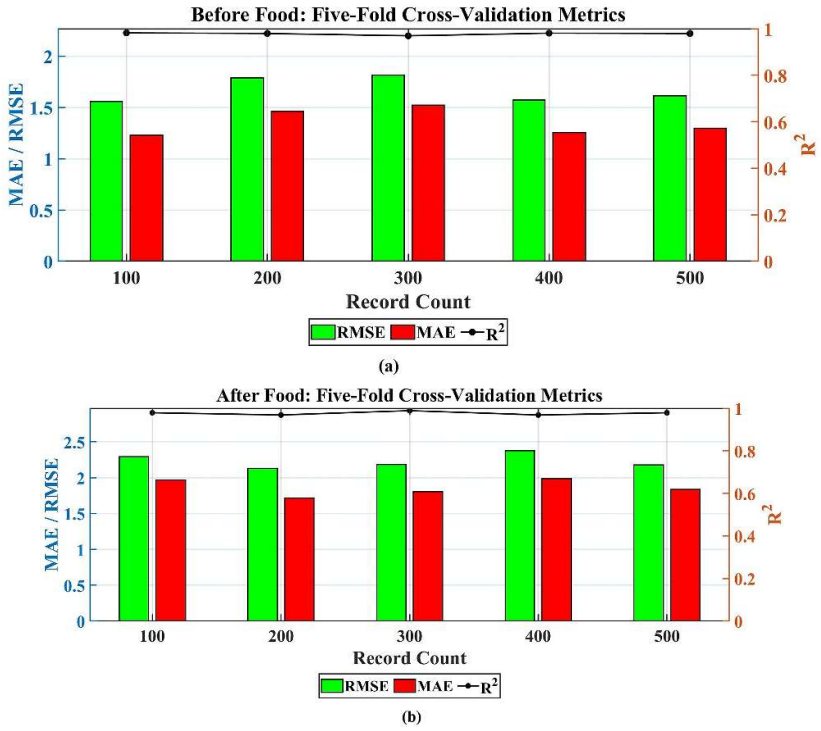
Comment.2	**The manuscript reports glucose prediction accuracies up to 98%, yet the machine-learning or regression framework is not sufficiently described or validated. Important evaluation metrics such as cross-validation strategy, patient-independent testing, Clarke Error Grid analysis, and long-term stability are missing.**	
Response.2	 **Fig.22:** Five fold CV prediction, **Fig.22 (a):** Before food five fold CV prediction, **Fig.22 (b):** After food five fold CV prediction  **Fig.23:** Five fold cross validation, **Fig.23 (a):** Before five fold cross validation, **Fig.23 (b):** After five fold cross validation	22 & 23

Fig.22 shows the fivefold Cross Validation (CV) prediction for before food prediction, Fig.22 (a) shows the before food fivefold CV after food prediction, Fig.22 (b) shows the after food fivefold CV prediction. Fig.23 shows the fivefold cross validation Fig.23 (a) shows the before food fivefold cross validation, Fig.23 (b) shows the after food fivefold cross validation. The fivefold cross validation shows strong predictive agreement before and after food intake through R2 values of 0.9787 and 0.9987 for fivefold cross validation.

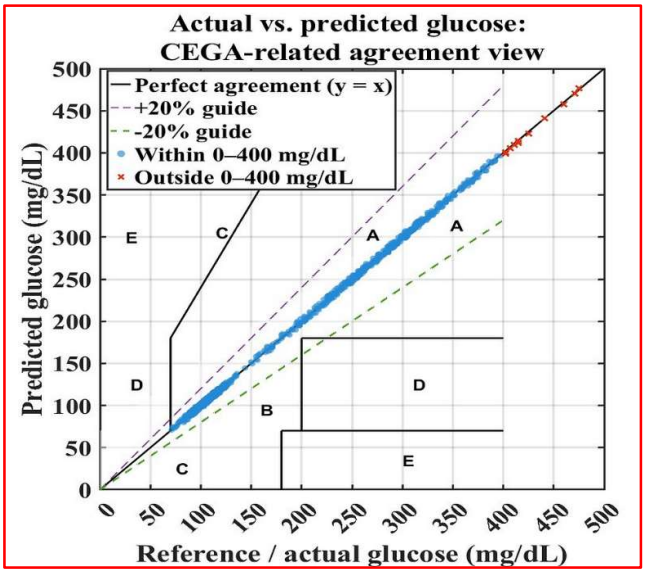


Fig.24: Clarke error grid analysis

Fig.24 shows the Clarke error grid analysis. Clarke Error Grid Analysis (CEGA) evaluates the clinical accuracy of predicted blood glucose levels compared with actual measurements. Most points fall within Zone A indicates accurate predictions.

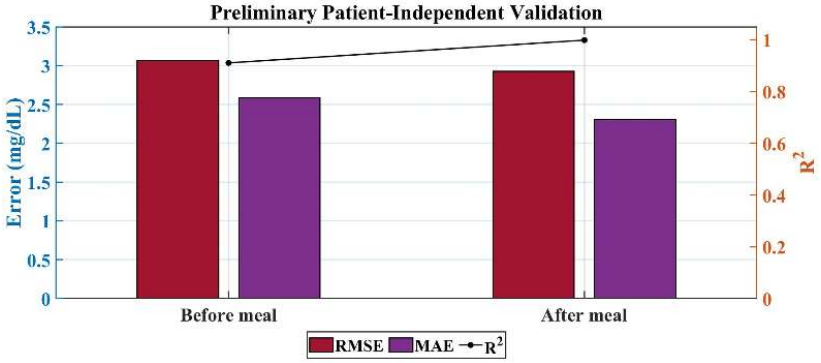


Fig.25: Preliminary patient independent validation

Fig.25 shows the preliminary patient independent validation. The patient independent validation demonstrates comparable predictive performance before and after meals. RMSE decreases from 3.1 to 2.9 mg/dL, MAE decreases from approximately 2.6 to 2.3 mg/dL which indicates reduced prediction errors after food. Coefficient of determination (R2) increases from 0.91 to 0.95, improves the goodness of fit under postprandial conditions. Table 22 shows the long-term stability of different patients.

Table 22: Long term stability of different patients

Patients ID	Gender/ Age	Glucose Test	Day and time	Date	Lab Value (mg/dL) @ Glucometer	TQWT Sub band_1	SAIH+ AHRF value (mg/dL)	Accuracy %
Patient 001	Male/ 47	Before Food	Day 1 & 6AM	10/11/2025	152	31.60	157.28	96.64
		After Food	Day 1 & 8AM	10/11/2025	209	30.98	212.52	98.34
		Random Test	Day 3 & 5PM	13/11/2025	161	30.05	164.45	97.90
		Random Test	Day 5 & 1PM	15/11/2025	95	19.20	97.73	97.21
		Random Test	Day 12& 4 PM	22/11/2025	75	14.85	78.96	94.98
Patient 002	Male/51	Before Food	Day 1 & 7AM	11/11/2025	148	29.99	151.93	97.41
		After Food	Day 1 & 9AM	11/11/2025	204	30.72	205.6	99.22
		Random Test	Day 2 & 2.30PM	12/11/2025	145	29.17	148.2	97.84
		Random Test	Day 42 & 11 AM	23/12/2025	106	21.62	109.55	96.76
Patient 003	Male/38	Before Food	Day 1 & 5AM	12/11/2025	154	27.69	162.44	94.80
		After Food	Day 1 & 7AM	12/11/2025	216	30.92	220.89	97.79
		Random Test	Day 3 & 1.30PM	15/11/2025	161	30.29	163.21	98.65
		Random Test	Day 8 & 10PM	20/11/2025	72	14.82	74.08	97.19
Patient 004	Female / 62	Before Food	Day 1 & 8AM	14/11/2025	99	19.95	105.63	93.72
		After Food	Day 1 & 10AM	14/11/2025	216	30.92	230.36	93.77
		Random Test	Day 2 & 9AM	15/11/2025	169	30.40	179.09	94.37
		Random Test	Day 3 & 3.30PM	17/11/2025	96	18.99	100.73	95.30
Patient 005	Female / 33	Before Food	Day 1 & 6AM	20/11/2025	232	31.48	243.12	95.43
		After Food	Day 1 & 8AM	20/11/2025	301	32.66	304.63	98.81
		Random Test	Day 10 & 8PM	30/11/2025	246	31.79	252.16	97.56
		Random Test	Day 32 & 10AM	22/12/2025	152	29.61	163.21	93.13

Comment.3	**The manuscript currently lacks sufficient comparison with recently reported microwave biomedical sensing systems. Without a more rigorous discussion against representative studies published in 10.1109/TMTT.2023.3324432, the actual novelty, structural innovation, and technical advancement of the proposed method remain insufficiently demonstrated.**	
Response.3	Microwave glucose-sensing systems sensitivity need to be improved through advanced resonant and multimode configurations. [50] reconfigurable multimode microwave sensor integrated with a controllable spoof surface plasmon polariton transmission line with a spoof localized surface plasmon multimode resonator. Resonance frequency and transmission-amplitude based sensor has achieved sensitivities of 9.86×10^{-2} MHz/(mg/dL) and 2×10^{-2} /(mg/dL), over a glucose concentration range of 0–800 mg/dL, with an approximately 0.99 correlation between the two sensing mechanisms. Proposed antenna operates at 4.2 GHz, with skin, fat, and muscle represented as separate layers. Table 21 shows the SAR values for a multilayer tissue phantom.	22
Comment.4	**The manuscript contains numerous overly strong claims regarding "clinical viability," "subclinical detection," and "real-time pancreas sensing," but these claims are not supported by clinical trials or medically validated datasets. The presented experiments appear limited to controlled laboratory conditions, which reduces the translational significance of the work.**	
Response.4	(Supplementary table ST6 shows Clinically glucose measured and TQWT Sub band_1 values obtained for the pancreas region using CGM).	

(Supplementary table ST6) Table: 6 Clinically glucose measured and TQWT Sub band_1 obtained for the pancreas region for CGM

Patient ID	Gender/ Age	Day and time	Actual Glucose Value (mg/dL)	Glucose Measurement Test	Measurement Condition	Activity/ Food Intake	TQWT Subband_1	Predicted glucose value (mg/dL)	Accuracy
P 001	Male / 43	Day 1/ 5.30 AM	195	Laboratory plasma glucose Test	Before Food	Jogging/ Fasting	30.83	191.56	98.24
		Day 1/ 8.30 AM	248	Laboratory plasma glucose Test	After Food	Millet rice	31.55	236.51	95.37
		Day 10/ 10.40 AM	166	Glucometer	Random Glucose Test	Vegetable	30.33	161.89	97.52
		Day 40/ 8.30 PM	170	Glucometer	Random Glucose Test	Egg curry + rice	30.34	169.52	99.72
		Day 48/ 12.30 PM	81	Glucometer	Random Glucose Test	Aval upma	16.66	78.91	97.42
P002	Female / 38	Day 2 / 6:00 AM	132	Laboratory plasma glucose test	Before food	Walking / Fasting	25.81	129.44	98.06
		Day 2 / 8:45 AM	186	Laboratory plasma glucose test	After food	Idli + sambar	28.26	180.72	97.16
		Day 12 / 11:15 AM	145	Glucometer	Random glucose test	Fruit	26.44	141.83	97.81
		Day 30 / 7:45 PM	158	Glucometer	Random glucose test	Chapati + dal	27.91	154.21	97.60
		Day 45 / 1:00 PM	98	Glucometer	Random glucose test	Vegetable upma	20.12	96.27	98.23

Patient ID	Gender/ Age	Day and time	Actual Glucose Value (mg/dL)	Glucose Measurement Test	Measurement Condition	Activity/ Food Intake	TQWT Subband_1	Predicted glucose value (mg/dL)	Accuracy
P003	Male / 51	Day 1 / 5:45 AM	142	Laboratory plasma glucose test	Before food	Fasting / Rest	26.11	138.76	97.72
		Day 1 / 9:00 AM	221	Laboratory plasma glucose test	After food	Rice + sambar	30.78	214.35	96.99
		Day 9 / 10:30 AM	178	Glucometer	Random glucose test	Walking	29.11	173.84	97.66
		Day 28 / 8:15 PM	164	Glucometer	Random glucose test	Dosa + chutney	28.45	160.92	98.12
		Day 44 / 12:15 PM	105	Glucometer	Random glucose test	Vegetable rice	22.17	102.61	97.72
P004	Female / 46	Day 3 / 6:15 AM	118	Laboratory plasma glucose test	Before food	Fasting	23.65	115.82	98.15
		Day 3 / 8:30 AM	172	Laboratory plasma glucose test	After food	Pongal	27.18	167.13	97.17
		Day 14 / 11:00 AM	126	Glucometer	Random glucose test	Household activity	24.57	123.48	98.00
		Day 32 / 7:30 PM	149	Glucometer	Random glucose test	Rice + vegetable curry	26.39	145.66	97.76
		Day 47 / 12:45 PM	92	Glucometer	Random glucose test	Idli	19.78	90.24	98.09

Patient ID	Gender/ Age	Day and time	Actual Glucose Value (mg/dL)	Glucose Measurement Test	Measurement Condition	Activity/ Food Intake	TQWT Subband_1	Predicted glucose value (mg/dL)	Accuracy
P005	Male / 35	Day 1 / 5:30 AM	108	Laboratory plasma glucose test	Before food	Fasting / Jogging	22.46	105.93	98.08
		Day 1 / 8:20 AM	154	Laboratory plasma glucose test	After food	Dosa + sambar	25.93	150.42	97.68
		Day 11 / 10:50 AM	137	Glucometer	Random glucose test	Cycling	25.12	133.68	97.58
		Day 29 / 8:00 PM	161	Glucometer	Random glucose test	Rice + egg	27.64	156.84	97.42
		Day 43 / 12:20 PM	89	Glucometer	Random glucose test	Vegetable salad	18.96	87.31	98.10
P006	Female / 57	Day 2 / 5:50 AM	154	Laboratory plasma glucose test	Before food	Fasting	27.03	149.83	97.29
		Day 2 / 8:40 AM	236	Laboratory plasma glucose test	After food	Rice + lentils	31.21	228.64	96.88
		Day 13 / 10:35 AM	193	Glucometer	Random glucose test	Light walking	30.02	187.45	97.12
		Day 31 / 8:10 PM	181	Glucometer	Random glucose test	Chapati + curry	29.54	176.32	97.41
		Day 46 / 12:10 PM	112	Glucometer	Random glucose test	Vegetable upma	22.84	109.76	98.00

Patient ID	Gender/ Age	Day and time	Actual Glucose Value (mg/dL)	Glucose Measurement Test	Measurement Condition	Activity/ Food Intake	TQWT Subband_1	Predicted glucose value (mg/dL)	Accuracy
P007	Male / 49	Day 1 / 6:10 AM	126	Laboratory plasma glucose test	Before food	Fasting	24.76	122.91	97.55
		Day 1 / 8:50 AM	198	Laboratory plasma glucose test	After food	Idli + chutney	29.41	192.17	97.06
		Day 10 / 11:20 AM	153	Glucometer	Random glucose test	Walking	27.19	149.72	97.86
		Day 34 / 7:50 PM	146	Glucometer	Random glucose test	Rice + fish curry	26.18	142.58	97.66
		Day 49 / 12:35 PM	94	Glucometer	Random glucose test	Millet upma	19.33	91.88	97.74
P008	Female / 42	Day 2 / 5:40 AM	101	Laboratory plasma glucose test	Before food	Fasting	21.94	99.16	98.18
		Day 2 / 8:35 AM	167	Laboratory plasma glucose test	After food	Pongal + sambar	26.78	162.92	97.56
		Day 15 / 10:45 AM	121	Glucometer	Random glucose test	Household activity	23.62	118.34	97.80
		Day 33 / 8:20 PM	139	Glucometer	Random glucose test	Chapati + vegetables	25.77	135.47	97.46
		Day 48 / 12:25 PM	86	Glucometer	Random glucose test	Fruit + nuts	18.21	84.61	98.38

Patient ID	Gender/ Age	Day and time	Actual Glucose Value (mg/dL)	Glucose Measurement Test	Measurement Condition	Activity/ Food Intake	TQWT Subband_1	Predicted glucose value (mg/dL)	Accuracy
P009	Male / 61	Day 1 / 6:05 AM	169	Laboratory plasma glucose test	Before food	Fasting	28.15	164.83	97.53
		Day 1 / 8:45 AM	254	Laboratory plasma glucose test	After food	Rice + curry	32.11	245.73	96.74
		Day 12 / 11:10 AM	204	Glucometer	Random glucose test	Walking	30.91	197.56	96.84
		Day 36 / 8:05 PM	188	Glucometer	Random glucose test	Dosa + chutney	30.15	182.91	97.29
		Day 50 / 12:40 PM	116	Glucometer	Random glucose test	Vegetable rice	23.14	112.84	97.28
P010	Female / 33	Day 3 / 5:55 AM	95	Laboratory plasma glucose test	Before food	Fasting	20.82	93.72	98.65
		Day 3 / 8:25 AM	143	Laboratory plasma glucose test	After food	Idli + sambar	24.66	139.84	97.79
		Day 16 / 10:55 AM	110	Glucometer	Random glucose test	Walking	22.17	107.61	97.83
		Day 38 / 7:40 PM	132	Glucometer	Random glucose test	Rice + vegetable curry	24.52	128.94	97.68
		Day 49 / 12:15 PM	83	Glucometer	Random glucose test	Aval upma	17.94	81.26	97.90